

Assessment Growth and Section Performance Analytics in R

Public-Safe BOY/EOY Improvement Study

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Recommendation

This study evaluates **beginning-of-year to end-of-year improvement** in a public-safe education assessment extract. The business question is: which course sections improved more or less than expected after accounting for starting performance, readiness, attendance, course track, grade level, and school-year context?

Use BOY/EOY gain as the headline metric, but use **adjusted growth signals** for section review. Raw gains are easy to understand, but they can reward sections that started low and penalize sections that started near a ceiling. The adjusted signal compares observed gain with expected gain for a similar starting profile.

The analysis includes 1,737 paired BOY/EOY records across 174 section-year groups and 5 simulated teachers. The average BOY score is 48.5, the average EOY score is 54.2, and the mean raw gain is 5.72 points.

The selected expected-growth model is **Readiness-augmented**. It achieved holdout RMSE **4.399** and holdout R-squared **0.181**. Section signals should be used for instructional review, curriculum support, and follow-up analysis; they should not be used as automatic teacher evaluation or personnel decisions.

Approach and Rationale

This study evaluates performance using **BOY/EOY score gain**: how much students improved from beginning of year to end of year. A raw gain ranking is easy to understand, but it can be misleading because sections with different starting levels, attendance patterns, and course contexts have different growth opportunities.

To create a fairer comparison, the model estimates each student's **expected BOY/EOY gain** from starting score, readiness, attendance, grade level, course context, and school-year timing. The adjusted gain signal is actual gain minus expected gain.

In plain terms, the report asks: did this section improve more or less than we would expect for students with similar starting profiles?

Direct Answers

1. The main metric is BOY/EOY score improvement: end-of-year score minus beginning-of-year score for the same simulated student in the same section and teacher context.
2. The average raw gain is 5.72 points across 1,737 paired records.
3. Raw section gains are reported, but the primary comparison is adjusted growth: observed section gain minus expected gain from the selected model.
4. The section review layer flags 8 section-year groups above expected growth and 5 below expected growth, with 156 within expected range. Start with the audit queue below rather than a raw ranking.
5. Course and teacher summaries are pattern-finding views. Alg 1 is the clearest course support review; Precalc is the clearest course bright spot; TCH-001 merits a composition-aware support review.

Initial audit queue

Level	Target	Reason
Section support	30-31 S09 / TCH-002 / Alg 2	Raw 2.33 vs expected 6.65; signal -2.05.
Section support	27-28 S08 / TCH-002 / Alg 2	Raw 4.04 vs expected 6.69; signal -1.50.
Section support	26-27 S03 / TCH-001 / Geometry	Raw 3.27 vs expected 6.45; signal -1.50.
Section bright spot	30-31 S06 / TCH-002 / Alg 2	Raw 10.39 vs expected 6.39; signal +1.89.
Section bright spot	30-31 S19 / TCH-005 / AP Calc AB	Raw 9.09 vs expected 6.13; signal +1.61.
Course support	Alg 1	Course signal -0.34; raw 6.60 vs expected 6.94.
Course bright spot	Precalc	Course signal +0.10; raw 5.73 vs expected 5.63.
Teacher support	TCH-001	Average signal -0.17 across 23 sections; review with composition context.

Data Audit

The analysis starts from a public-safe assessment extract. A record enters the growth model only when the same simulated student has valid BOY and EOY scores in the same section and with the same simulated teacher. This keeps the improvement metric tied to a section experience instead of mixing students across sections.

Measure	Value
Raw assessment rows	4,018
BOY/EOY candidate pairs	2,009
Included paired records	1,737
Unique public-safe student IDs	671
Unique section-year groups	174
Unique simulated teachers	5
Mean BOY score	48.5
Mean EOY score	54.2
Mean BOY/EOY gain	5.7
Median section paired records	10

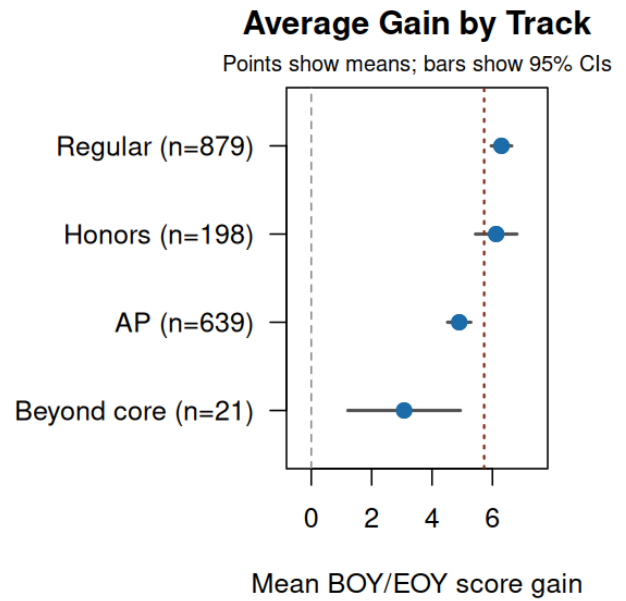
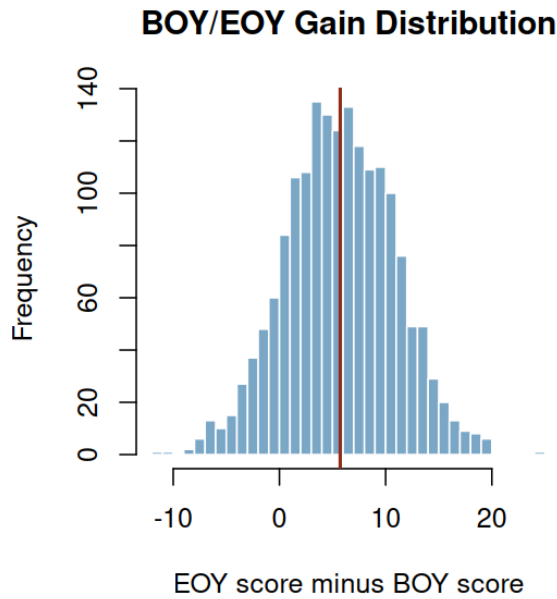
The extract uses simulated identifiers and generalized score/readiness behavior from a bootstrapped assessment workflow. It is not a release of real student records or real personnel data.

Raw Section Improvement

The first layer is descriptive: calculate the BOY/EOY score gain inside each section-year group and run a paired-improvement t-test against zero. This answers whether a section improved, but it does not by itself prove that the section improved more than expected given its starting point.

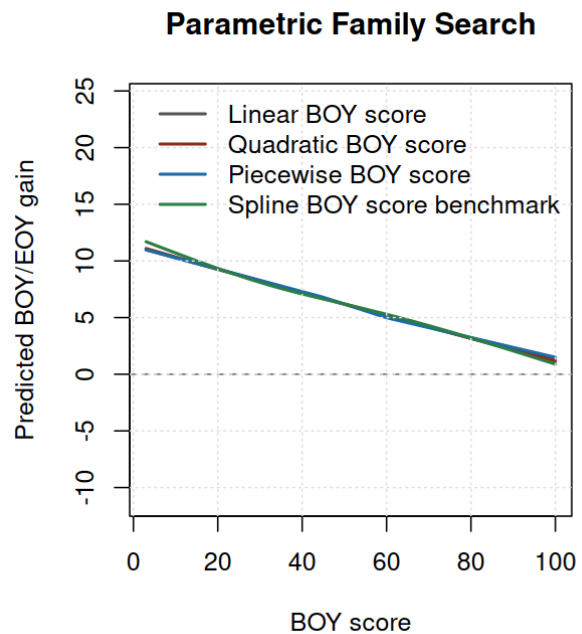
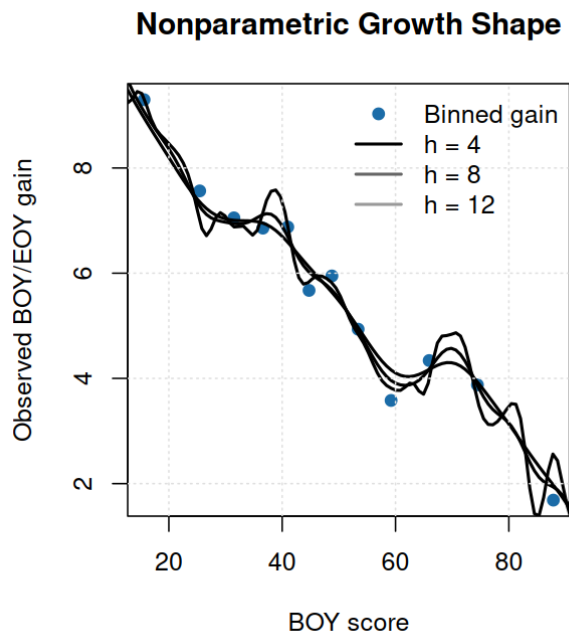
The table below shows high-signal section-year groups from the review layer, with their raw BOY/EOY t-test results included for context. The full section t-test table is generated as [reports/section_ttests.csv](#).

Section	N	BOY	EOY	Gain	95% CI	p-value
30-31 S06	9	42.4	52.8	10.39	7.18 to 13.60	<0.001
30-31 S19	12	37.6	46.7	9.09	6.66 to 11.52	<0.001
27-28 S17	11	43.9	52.6	8.79	5.37 to 12.22	<0.001
29-30 S17	11	37.1	45.9	8.77	5.77 to 11.78	<0.001
30-31 S08	11	35.0	44.0	9.07	4.80 to 13.34	<0.001
31-32 S02	10	40.1	49.1	9.02	6.32 to 11.73	<0.001
29-30 S03	12	41.0	49.7	8.73	6.24 to 11.22	<0.001
26-27 S06	9	30.0	39.9	9.85	6.41 to 13.28	<0.001



Adjusted Growth Model

The adjusted model estimates expected BOY/EOY gain from starting score/readiness and context. This is the key step that makes the analysis more useful than a raw gain ranking: it accounts for floor effects, ceiling effects, attendance context, course track, grade level, and school-year timing.

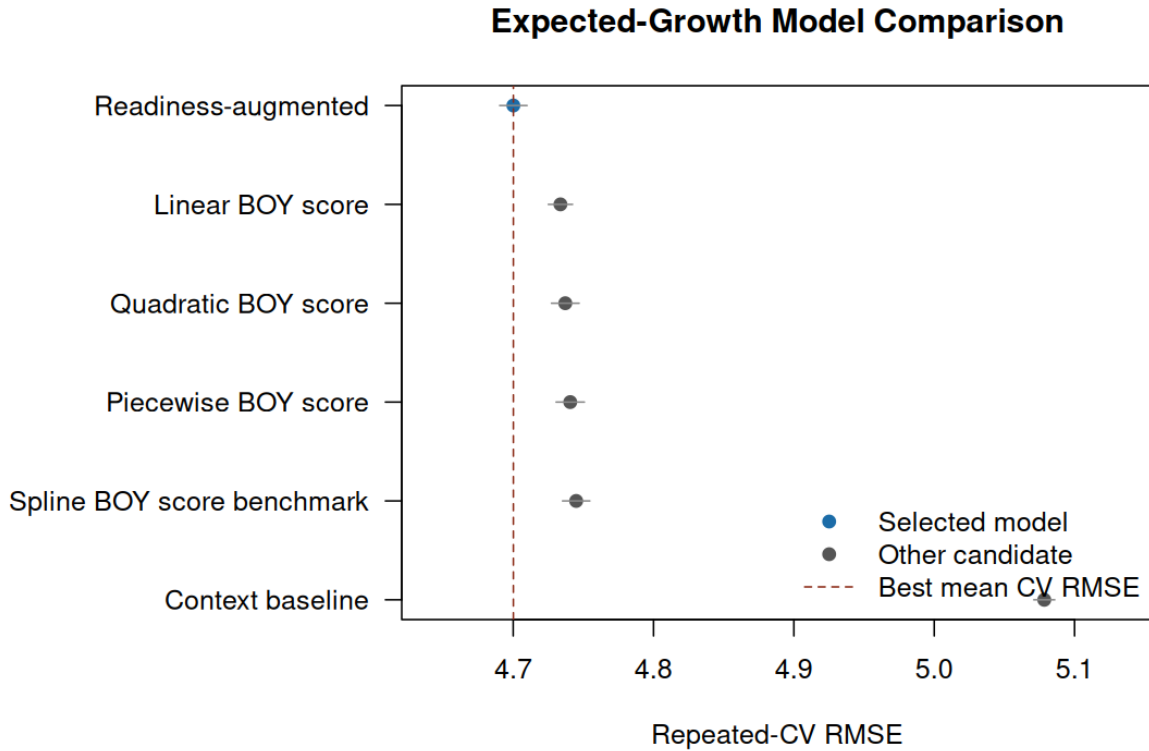


Model family review

Family	Decision	CV RMSE	Holdout RMSE
Context baseline	Baseline comparator.	5.078	4.802
Linear BOY score	Useful simple challenger.	4.734	4.428
Quadratic BOY score	Compared for nonlinear gain shape.	4.737	4.427
Piecewise BOY score	Interpretable nonlinear challenger.	4.741	4.432
Readiness-augmented	Selected operating model.	4.700	4.399
Spline BOY score benchmark	Benchmark only; not selected unless it materially improves validation.	4.745	4.422

Model	Selected	CV RMSE	Holdout RMSE	Holdout R2
Readiness-augmented	Yes	4.700	4.399	0.181
Linear BOY score		4.734	4.428	0.170
Quadratic BOY score		4.737	4.427	0.171
Piecewise BOY score		4.741	4.432	0.169
Spline BOY score benchmark		4.745	4.422	0.172
Context baseline		5.078	4.802	0.024

The full model-comparison table is generated as [reports/growth_model_comparison_display.csv](#).

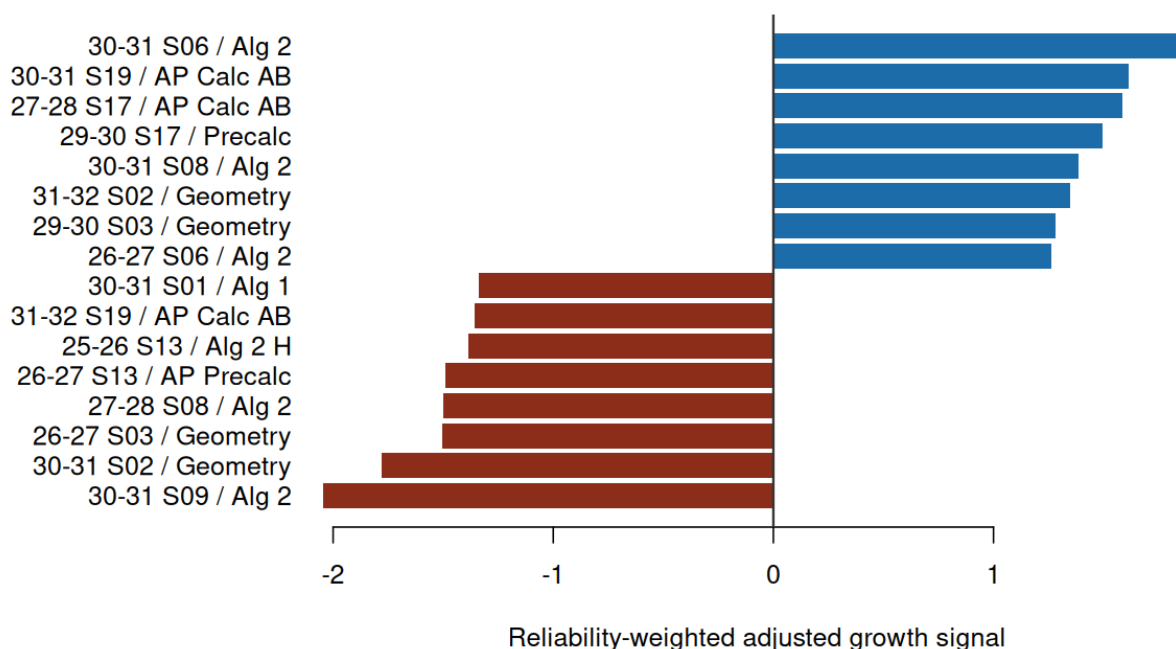


Section Performance Signals

For each section-year group, the adjusted signal is the reliability-weighted average residual: observed gain minus expected gain, weighted toward zero for smaller groups. Positive values mean the section improved more than expected for its starting mix; negative values mean it improved less than expected.

Section	Teacher	N	Raw	Expected	Signal	Result
30-31 S06	TCH-002	9	10.39	6.39	1.89	Above
30-31 S19	TCH-005	12	9.09	6.13	1.61	Above
27-28 S17	TCH-005	11	8.79	5.77	1.59	In range
29-30 S17	TCH-003	11	8.77	5.92	1.50	Above
30-31 S08	TCH-002	11	9.07	6.42	1.39	In range
31-32 S02	TCH-002	10	9.02	6.33	1.35	In range
29-30 S03	TCH-002	12	8.73	6.38	1.28	Above
26-27 S06	TCH-002	9	9.85	7.18	1.26	In range
30-31 S09	TCH-002	9	2.33	6.65	-2.05	Below
30-31 S02	TCH-002	12	2.22	5.48	-1.78	In range
26-27 S03	TCH-001	9	3.27	6.45	-1.50	Below
27-28 S08	TCH-002	13	4.04	6.69	-1.50	Below
26-27 S13	TCH-004	10	2.27	5.25	-1.49	Below
25-26 S13	TCH-003	11	3.99	6.64	-1.39	In range
31-32 S19	TCH-005	12	2.41	4.90	-1.36	In range
30-31 S01	TCH-001	10	3.28	5.96	-1.34	Below

Sections Farthest Above or Below Expected Growth



The full section signal table is generated as [reports/section_adjusted_signals.csv](#) so reviewers can inspect all section-year groups, not only the highlights shown in the report.

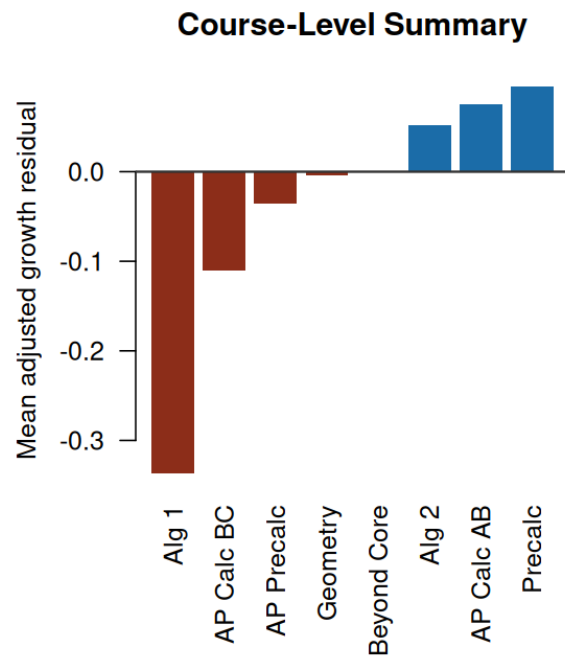
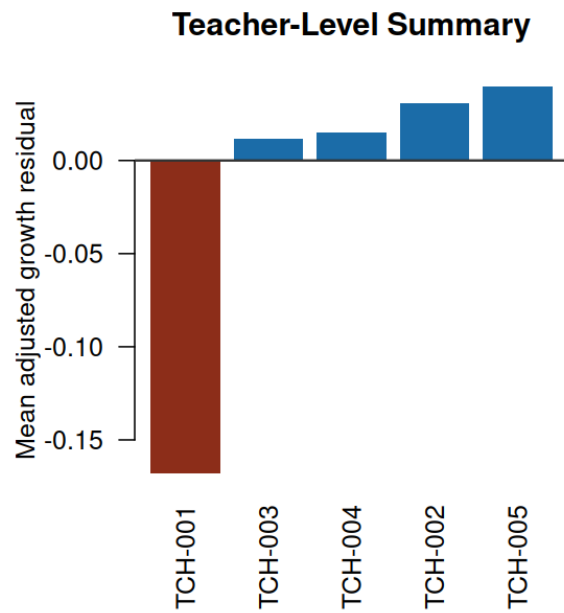
Instructor and Course Summary

Teacher and course summaries aggregate the section-level evidence. They are useful for spotting patterns that deserve follow-up, such as a course sequence that may need curriculum review or a teacher group whose sections repeatedly exceed expected growth. They should not be treated as standalone teacher quality scores.

Teacher	Sections	Records	BOY	EOY	Raw gain	Expected gain	Adjusted signal
TCH-005	41	420	54.6	59.4	4.81	4.77	0.04
TCH-002	40	424	39.6	46.1	6.48	6.45	0.03
TCH-004	36	313	55.8	60.9	5.11	5.10	0.01
TCH-003	34	351	50.6	56.5	5.95	5.94	0.01
TCH-001	23	229	40.7	47.2	6.49	6.66	-0.17

Course-level summary

Course	Sections	Records	Raw	Expected	Signal
Precalc	21	216	5.73	5.63	0.10
AP Calc AB	27	294	5.21	5.14	0.07
Alg 2	23	236	6.57	6.52	0.05
Beyond Core	6	21	3.08	3.08	0.00
Alg 2 H	20	198	6.12	6.12	0.00
Geometry	31	334	6.39	6.40	-0.00
AP Precalc	22	214	5.10	5.14	-0.04
AP Calc BC	14	131	3.86	3.97	-0.11
Alg 1	10	93	6.60	6.94	-0.34

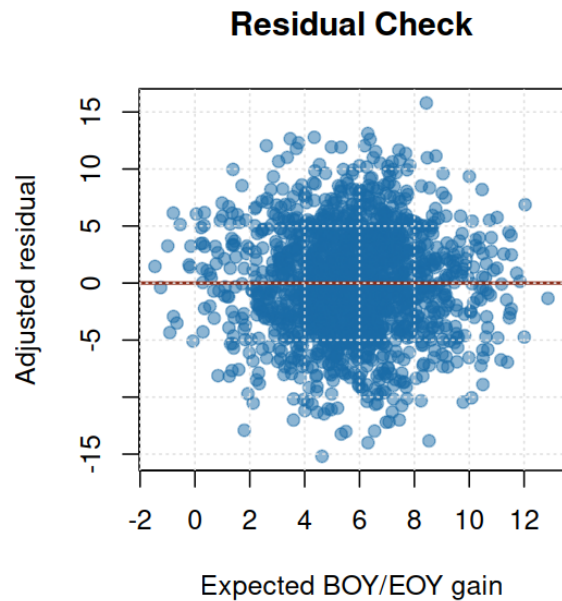
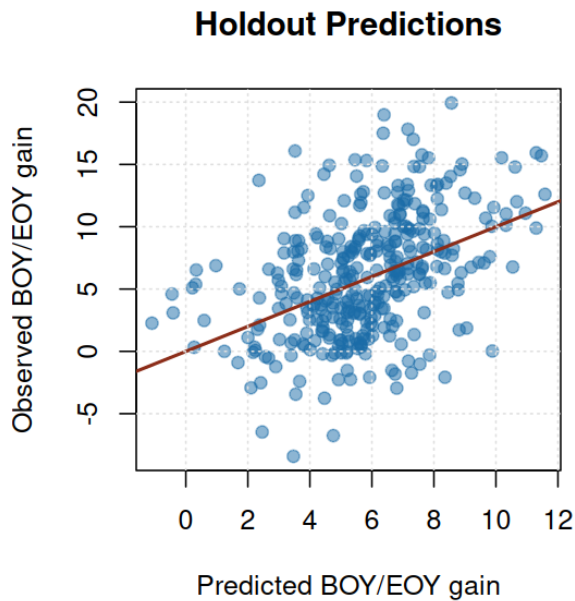


Diagnostics and Sensitivity

The diagnostics below check whether the expected-growth model is centered and whether the raw section rankings materially differ from adjusted rankings.

Diagnostic	Estimate	Interpretation
Holdout RMSE	4.399	Typical holdout prediction error on BOY/EOY gain
Holdout MAE	3.512	Average absolute holdout prediction error
Holdout R-squared	0.181	Share of holdout gain variation explained by the model
Residual mean, all pairs	0.000	Near 0 means expected gain is centered overall
Residual SD, all pairs	4.599	Residual spread used to judge section signal uncertainty

Measure	Value
Included paired records	1,737
Section-year groups	174
Groups with at least 5 paired records	169
Groups with at least 8 paired records	159
Mean raw BOY/EOY gain	5.72
Raw-vs-adjusted rank correlation	0.797
Top-10 overlap, raw vs adjusted ranking	70.0%



Bottom Line

- BOY/EOY improvement is the right stakeholder metric because it is easy to explain and aligned with instructional growth.
- Raw improvement should be shown, but adjusted growth should drive section review because starting level, attendance, course track, and ceiling effects matter.
- The strongest use case is instructional review: identify sections that exceed expected growth, sections that lag expected growth, and course patterns that deserve support.
- Avoid framing results as teacher quality rankings. The outputs are public-safe analytical signals, not personnel decisions.

Reproducibility

Rebuild the full evidence packet with `make all`. The core pipeline uses base R, the included public-safe extract, and no credentials, private files, or network access.

Public-Safety Statement

This report is an original public-safe portfolio artifact. It excludes private coursework prompts, exams, rubrics, syllabi, lecture transcripts, source datasets, personal data, real student-identifiable records, real personnel records, credentials, and copyrighted source documents.